Research Square

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Individualized Training Based on the Force-Velocity Profile: A Systematic Review with Meta-Analysis examining the Effects on Motor Performance

Christofer Wolte

christofer.wolte@aau.at

University of Klagenfurt https://orcid.org/0009-0005-0466-9830

Thomas Gronwald

Medical School Hamburg

Marcelle Schaffarczyk

Medical School Hamburg

Jan Wilke

University of Bayreuth

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Abstract

Background

Exercise has numerous benefits for health, well-being and performance. However, due to factors such as genetics or training status, the individual response can be highly different. Force-velocity (FV) based training is a popular method to individualize exercise programs aiming to improve speed and power. This systematic review investigated the effects of FV based training on motor performance.

Methods

A systematic literature search was conducted by two independent examiners using PubMed, Web of Science, and Google Scholar. We included randomized controlled trials involving healthy adults and comparing individualized (FV) to non-individualized training programs with a minimal duration of four weeks. Study quality was evaluated using the PEDro scale, publication bias was checked by inspection of funnel plots. We used robust variance estimation to pool the effects of individualized vs. non-individualized training for sprint time, strength, and jump height.

Results

Searches returned 684 articles, and n = 10 papers were included. Study quality was good (5.3 ± 0.8 / 7 points on the PEDro scale) and no indication of publication bias was found. Meta-analysis did not reveal differences between FV based and non-individualized training for strength (SMD: -0.04, 95%CI: -0.34 to 0.26, p = 0.72, I2: 0%), sprint time (SMD: 0.28, 95%CI: -0.75 to 1.32, p = 0.49, I2: 69,7%), and jump height (SMD: 1.8, 95%CI: -0.57 to 4.2, p = 0.11, I2: 90.8%).

Conclusion

Although FV profiling represents a plausible approach to individualize speed and power training, our meta-analysis does not support its application for performance reasons at present. Future research should investigate more specific conditions and homogenous populations such as elite athletes.

Background

It is well established that exercise has a variety of benefits, increasing motor performance, preventing musculoskeletal injury, reducing the risk of non-communicable diseases, and improving mental health [1–7]. However, the response to training can be highly variable. Hubal et al. [8] submitted 585 participants to a 12-week progressive resistance training program [9]. Although the mean improvement in the 1-Repetition Maximum (1RM) was 54%, individual changes ranged from 0 to + 250%. Similarly, pre-post differences in maximal voluntary contraction force varied from − 32 to + 149%. High inter-individual

variability has also been observed in the adaptation to endurance-type exercise [10]. These examples illustrate that some participants may respond strongly to training programs, while others are poor or non-responders.

The factors contributing to variability in the training response have been the subject of contentious discussion. In 1999, Bouchard [11] investigated the role of genetics in the adaptation to chronic endurance training. A total of 481 sedentary individuals from 98 families exercised on a cycle ergometer for 20 weeks. Interestingly, the variance of changes in maximal oxygen uptake (VO2max) was 2.5 times higher between families than within families. Although this may indicate that the responsiveness to exercise is heritable to a certain extent, Mann and colleges (2014) described additional factors potentially affecting the training response. These included the homeostatic stress produced by the training, recovery and readiness to exercise, and nutrition [12].

In view of the substantial inter-individual differences in exercise effects and the numerous influencing factors, researchers have proposed that training should be individualized, as implied by the principle of individualization [13]. Indeed, the development of individualized training methods and programs has experienced a recent surge in popularity [14]. Briefly, the idea of related approaches is to tailor training parameters, such as intensity, volume, or exercise selection, to the individual's specific needs and characteristics (e.g., age, sex, recovery status, fitness level) [15]. This personalized approach is believed to achieve superior results compared to non-individualized training programs.

Among the many methods suggested to individualize training, force-velocity (FV) profiling has received particular attention [16][17]. It is based on the assumption that speed or jump height are dependent on both, force and movement velocity [18]. To determine the FV profile, the outcome of interest (e.g., sprint speed or jump height) is measured repeatedly but with increasing resistance (e.g., unloaded sprint, sprint with a sledge at increasing weights). From the generated data, a FV profile is constructed plotting force against velocity and using linear regression. The comparison of the obtained and optimal FV profiles reveals a force deficit, velocity deficit, or balance between both components. In case of a deficit, exercises selected to improve speed or jump height should address the FV imbalance. For instance, FV based training in individuals with a velocity deficit would focus on movement speed while athletes with a force deficit would be trained towards improving maximal power output.

Although several studies have examined the effectiveness of individualized training based on the FV profile, there is no systematic evaluation of the available evidence. Our systematic review with meta-analysis aimed to provide an overview of the current literature, examining the hypothesis that individualization of exercise training programs based on the FV profile is superior to non-individualized training regarding markers of speed and power.

Methods

A systematic review with meta-analysis was performed adhering to the PRIMSA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [19]. The study followed the

recommendations for ethical publishing of systematic reviews proposed by Wagner and Wiffen [20] and was prospectively registered in the PROSPERO database (CRD42024501418).

Search strategy and inclusion criteria

Two independent examiners (CW and MS) conducted a systematic literature search in PubMed and Web of Science. The search term was “(individual* OR personal* OR optimiz*) AND force-velocity AND (train* OR exercise)”. In addition to the systematic search, an additional search was performed in Google Scholar, screening the first 100 hits produced by the same search term [21, 22] [23]. Furthermore, references from all included studies were checked [24].

Studies were included when (1) having a randomized controlled trial design (RCT), (2) investigating healthy adults, (3) applying an individualized training intervention (minimum 4 weeks) based on the force-velocity profile, and (4) having a control group with an active but non-individualized training program. Any studies involving patients with chronic diseases or individuals experiencing pain, as well as studies using a crossover design and including children and adolescents were excluded.

Methodological quality, reporting bias and certainty about the evidence

The Physiotherapy Evidence Database (PEDro) scale [25] was used by two independent reviewers (TG and MS) in order to assess the methodological quality of the articles included. The PEDro scale consists of 11 different items related to scientific rigor; item 1 is captures external validity (yes/no), items 2 to 11 internal validity. The items are rated using 0 as absent or unclear, and 1 as present. The highest score in the PEDro scale is 10 because item 1 is not included in the total score (lowest score: 0). Given that the assessors are rarely blinded, and that it is impossible to blind the participants and investigators in supervised exercise interventions, the items 5–7 related to blinding were removed from the total score for our study [26]. The highest score, therefore, was 7 points. To facilitate interpretation, the total score can be interpreted as follows: 6 to 7 = “excellent“, 5 = “good“, 4 = “moderate“, and 0 to 3 = “poor“.

To evaluate the risk of a reporting bias, we constructed and visually inspected funnel plots of standard error and effect size. To assess the certainty about the evidence, we used the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) method for each individual outcome [27]. Due to the randomized controlled trial design, ratings started with the outcome (strength, jump height, sprint time) being classified as ‘high certainty about the evidence’. In case of a) risk of bias (PEDro score < 4), b) imprecision (wide 95% CI), c) inconsistency (high heterogeneity), d) indirectness of evidence (measured outcome does not represent suggested construct), and e) publication bias (funnel plot asymmetry), certainty was rated down by one category for each weakness (e.g., from high to moderate). Quality was rated up by one category in case of a large-magnitude effects or dose-response gradients. This resulted in final certainty rating being high, moderate, low, or very low.

Data extraction

The following data were extracted by two examiners (CW and MS): sample size, participant characteristics, intervention details, measured outcomes and results (pre and post means and SD as well as pre-post changes and SD per intervention arm). The outcomes of the meta-analyses encompassed measures of strength (e.g., maximum voluntary force), jump height (counter-movement jump/CMJ, squat jump/SJ, and both pooled), and sprint time (e.g., 5m, 10m, 20m, 30m). Biomechanical information was gathered from each study. If a study assessed more than one parameter of the same outcome (e.g., 10 and 20 m sprint, or CMJ and SJ height), all effect sizes (ES) were extracted.

Data processing and statistics

Per trial arm, changes from pre to post were computed as $Mean_{post} - Mean_{pre}$ and standard deviations (SD) were pooled as

$$SD_{pooled} = \sqrt{\frac{(n_1 - 1) * SD_1^2 + (n_2 - 1) * SD_2^2}{(n_1 - 1) + (n_2 - 1)}}$$

In addition to pre-post differences, differences between groups and related SD were calculated. The authors of the primary studies were contacted if data (e.g., SD of changes) were missing and could not be read from figures. To account for within-study outcome dependency resulting from multiple outcomes (e.g., 1RM of squat and leg press), we used robust variance estimation to pool the effect sizes [28]. Standardized mean differences (SMD) and 95% confidence intervals (CI) were calculated for measures of strength, jump height, and sprint time. We used R (R Foundation for Statistical Computing, Vienna, Austria) with the robumeta (version 2.0, Fisher & Tipton, 2015) and meta packages. Effect sizes (ES) were interpreted as $0 \leq d < 0.2$ trivial, $0.2 \leq d < 0.5$ small, $0.5 \leq d < 0.8$ moderate, or $d \geq 0.8$ large [29]. Heterogeneity was quantified as τ^2. P values ≤ 0.05 were considered as statistically significant.

Results

Search results and study characteristics

The search returned a total of 684 articles (Fig. 1). After removal duplicates and application of the exclusion criteria, n = 10 studies were found eligible for inclusion.

The 10 papers [16, 17, 30–37] collectively evaluated 417 participants (120 men, 198 women; no sex reported by one study [33]) with a mean age of 29 ± 17 years. While 6 of the included studies examined the effect of force-velocity training (FVT) on jump capacities, 5 focused on sprint time, and 6 assessed

strength. The training duration ranged from 8 to 12 weeks and training frequency varied between two and three sessions per week (Table 1).

Table 1
Characteristics of the studies included into quantitative synthesis.

Study	Sample	Performance Level	Individualization parameter	Intervention Duration	Outcome
Lindberg et al. (2021)	40 male handball, ice hockey, and football players; 20 ± 4 years;	Professional (national team)	Theoretical SJ FV profile was calculated using SJ with five different weights (0, 20, 40, 60, and 80 kg).	10 weeks	10 and 30 m sprints, SJ and CMJ, 1 repetition maximum (1RM) squat and a leg press performance test
Escobar Álvarez et al. (2019)	46 female ballet dancers; 18,9 years	Not reported	Multiple series of loaded vertical jumps, CMJ.	9 weeks	CMJ
Barrera-Domínguez et al. (2023)	30 male basketball players; 22,8 ± 5,68 years	Not reported	CMJ, unilateral drop jumps, triple hop test, sprint, and change of direction.	8 weeks	Jumping ability CMJ, Drop Jump, sprint 10m, change of direction (COD)
Rodriguez-Lopez et al. (2021)	45 men and women; ≥ 65 years old	Not reported	FV used to determine 1-RM, F0, V0, and Pmax. Power-oriented resistance training with either light (LL-PT, 40% 1-RM) or heavy (HL-PT, 80% 1-RM) weight on leg press.	12 weeks	1-RM, F0 (force at zero velocity), V0 (maximum unloaded velocity), Pmax (maximum muscle force), RFD), muscle activation, muscle cross-sectional area (CSA), pennation angle (muscle architecture)
Jiménez-Reyes et al. (2017)	84 male football and rugby players; 23,1 ± 4,4 years	Semi-professional	FV imbalance: focused on the FV curve range, where both high loads for strength exercises (> 80% of a repetition maximum in squats) and speed exercises (with bodyweight at high speed) were used.	9 weeks	Jumping performance (unloaded SJ height)

SJ = squat jump, CMJ = counter-movement jump, FV = force-velocity, 1RM = one repetition maximum, CG = control group, IG = intervention group,

Study	Sample	Performance Level	Individualization parameter	Intervention Duration	Outcome
Zabaloy et al. (2020)	34 male rugby players; 22,17 ± 3,32 years	Professional	Not reported	7 weeks	Sprint performance, 1RM-SQ strength (maximum squat), jump height, (CMJ SJ), Pmax
Rakovic et al. (2018)	17 female handball players; IG = 23 ± 3 years, CG 23 ± 3 years	Semi-professional /professional	Not reported	8 weeks	Changes in maximum velocity (V0) and improvements in 0–30 m sprint time
Lindberg et al. (2022)	56 healthy males; 68 ± 5 years	Recreational	FV profile determined by leg press.	10 weeks	Stair climb, loaded stair climb, sit to stand, grip strength
Cross et al. (2018)	15 male soccer players (27 ± 5 years) and 21 (12 females, 9 males) rugby players (27 ± 2 years)	Semi-professional/ professional	FV profile determined by sprint.	10 weeks	Sprint performance 30-m unresisted, 30-m at 25%; 20-m at 50%; 20-m at 75%; 15-m at 100% BM or its´ 1080 Sprint equivalents
Simpson et al. (2021)	29 male rugby players; 24 ± 3 years	Professional	Not reported	8 weeks	10-m and 20-m sprint, squats with 3 repetitions, SJ under five load conditions
SJ = squat jump, CMJ = counter-movement jump, FV = force-velocity, 1RM = one repetition maximum, CG = control group, IG = intervention group,					

Risk of bias

Ratings of study qualities revealed a low risk of bias for all papers as the mean PEDro score was 5.3 ± 0.8, with values ranging from 4 to 6 (Table 2). Visual inspection of funnel plots suggested absence of a reporting bias for strength level and speed while some asymmetry (i.e., lack of small studies with negative effect) was identified for jump height (Fig. 3).

Quantitative synthesis

FVT was not superior to non-individualized training with regard to all tested parameters as no differences to control were found for strength (SMD: -0.04, 95% CI: -0.34 to 0.26, p = 0.72 I^2: 0%, 6 studies, 14 effect sizes), sprint time (SMD: 0.28, 95% CI: -0.75 to 1.32, p = 0.49, I^2: 70%, 5 studies, 22 effect sizes), general jump height (SMD: 1.8, 95% CI: -0.57 to 4.16, p = 0.11, I^2: 90%, 6 studies, 14 effect sizes), CMJ height (SMD: 0.96, 95% CI: -0.49 to 2.41, p = 0.13, Tau^2: 0.48, 4 studies, 6 effect sizes), and SJ height (SMD: 0.12, 95% CI: -0.67 to 0.92, p = 0.58, Tau^2: 0, 3 studies, 5 effect sizes). The certainty about the evidence was high for strength, low for sprint time, and very low for jump height.

Table 2
PEDro ratings of the included studies.

	Item score											
Study	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**Total**
Lindberg et al. (2021)	No	1	0	1	0	0	0	1	1	1	1	6
Lindberg et al. (2022)	No	1	0	0	0	0	0	1	1	1	1	5
Jiménez-Reyes et al. (2017)	No	0	0	0	0	0	0	1	1	1	1	4
Cross et al. (2018)	Yes	1	0	1	0	0	0	1	1	1	1	6
Rakovic et al. (2018)	Yes	1	0	1	0	0	0	1	1	1	1	6
Rodriguez-Lopez et al. (2021)	Yes	1	0	1	0	0	0	1	1	1	1	6
Simpson et al. (2021)	No	0	0	0	0	0	0	1	1	1	1	4
Zabaloy et al. (2020)	No	1	0	0	0	0	0	1	1	1	1	5
Escobar Àlvarez et al. (2020)	No	1	0	0	0	0	0	1	1	1	1	5
Barrera-Domínguez et al. (2023)	No	1	0	1	0	0	0	1	1	1	1	6

1 = eligibility criteria specified; 2 = subjects randomly allocated; 3 = concealed allocation; 4 = groups similar at baseline regarding important prognostic indicators; 5 = subject blinding; 6 = therapist blinding; 7 = assessor blinding; 8 = measures of 1 key outcome from 85% of subjects initially allocated to groups; 9 = subjects for whom outcome measures were available received treatment/control as allocated or data analyzed “intention to treat”; 10 = between-group statistical comparisons reported; 11 = point measures and measures of variability

Discussion

Individualization plays an instrumental role in the design of training programs, which is mainly due to the variability in the exercise response. Non-modifiable factors such as genomic predictors [38] or the influence of sex hormones [39] have been proposed to explain different adaptations to the same intervention [12]. In addition, Pickering et al. [40] suggested that considering training variables including volume, duration, and intensity may help to reduce heterogeneity. This systematic review investigated the effects of training based on the FV profile, a highly popular approach for exercise-based individualization of interventions aiming to improve speed and power [35] [37]. Of note, our meta-analysis of the available literature did not show superiority of FV based training over non-individualized training regimes.

The lack of FV based training effects partly contrasts with previous research on methods for exercise-based individualization. Evaluating a total of 58 original studies, Fleckstein et al. [41] found personalized training to slightly better combat chronic-nonspecific low back pain than standard training. Other recent reviews examining individualization [42] [43]) focused on exercise prescription based on heart rate variability (HRV). The authors reported improvements in submaximal physiological parameters, a lower number of non-responders, as well as a higher number of high responders with HRV-based training when compared to predefined training regimes. Regarding performance, however, effects seemed small or of questionable relevance.

Non-superiority of FV based individualization may be attributed to several aspects. First, an athlete's FV profile is significantly influenced by numerous factors such as body mass, joint range of motion, and gravity or friction [44]. These variables interact in a non-linear way, impacting performance differently and potentially limiting the effectiveness of purely FV based individualization approaches. Additionally, methodological differences in the individual studies (e.g., variation in measurement tools such as smartphone apps or linear transducers) and inadequate adjustments to the specific conditions (e.g., ground) may have impacted the results [44]. In sum, individualization may need to go beyond FV profiling and include a more comprehensive consideration of both personal and environmental constraints.

A second issue does not relate to the individualization method, but the available evidence itself. It needs to be acknowledged that the number of RCTs on FV based training is still small and that the certainty about the evidence of the present meta-analysis was low and very low for sprint time and jump height, respectively. The populations of the primary studies included healthy older persons and athletes, as well as individuals from a number of different sports such as handball, basketball, football, rugby, ice hockey, and ballet dancing. The latter is noteworthy because the nature and magnitude of FV imbalance could be sports specific. For instance, Escobar Álvarez et al. [32] found a large force deficit of more than 40% in both, their experimental and control groups. Against this background, using more homogenous populations in upcoming trials may help to better delineate the potential of FV based training. Finally, it may be that the potential of FV based training is direction dependent. While effect sizes for sprint time and strength were small or trivial, we found a non-significant, but large effect size for jump height. In view of the comparatively small number of available studies, additional research may help clarify if FV based training is only effective in jumping and/or other vertically directed movements.

Conclusion

Although FV profiling represents a plausible approach to individualize speed and power training, our meta-analysis does not support its application for performance optimization (particularly regarding strength and sprint time) at present. Future research should investigate more specific conditions and homogenous populations such as elite athletes.

Abbreviations

FV
Force–Velocity
1RM
1–Repetition Maximum
CMJ
Counter–Movement Jump
SJ
Squat Jump
PEDro
Physiotherapy Evidence Database
RCT
Randomized Controlled Trial
CI
Confidence Interval
SMD
Standardized Mean Difference
I^2
I–squared (a statistic measuring heterogeneity)
Pmax
Maximum Power Output
F0
Force at Zero Velocity
V0
Maximum Unloaded Velocity
RFD
Rate of Force Development
CSA
Cross–Sectional Area
GRADE
Grading of Recommendations Assessment, Development, and Evaluation
PRISMA
Preferred Reporting Items for Systematic Reviews and Meta–Analyses

PROSPERO
International Prospective Register of Systematic Reviews
IG
Intervention Group
CG
Control Group
COD
Change of Direction
BM
Body Mass
HRV
Heart Rate Variability
LL
PT–Light Load Power–Oriented Training
HL
PT–Heavy Load Power–Oriented Training

Declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and material

The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request. Data are located in controlled access data storage at University of Klagenfurt.

Competing interests

The authors have no relevant financial or non-financial interests to disclose.

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Author contributions:

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by MS, TG and CW. The first draft of the manuscript was written by CW and JW. JW had the idea for the article. CW and MS performed literature search. JW and CW made the analyses and drafted manuscript, all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Acknowledgments

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Figures

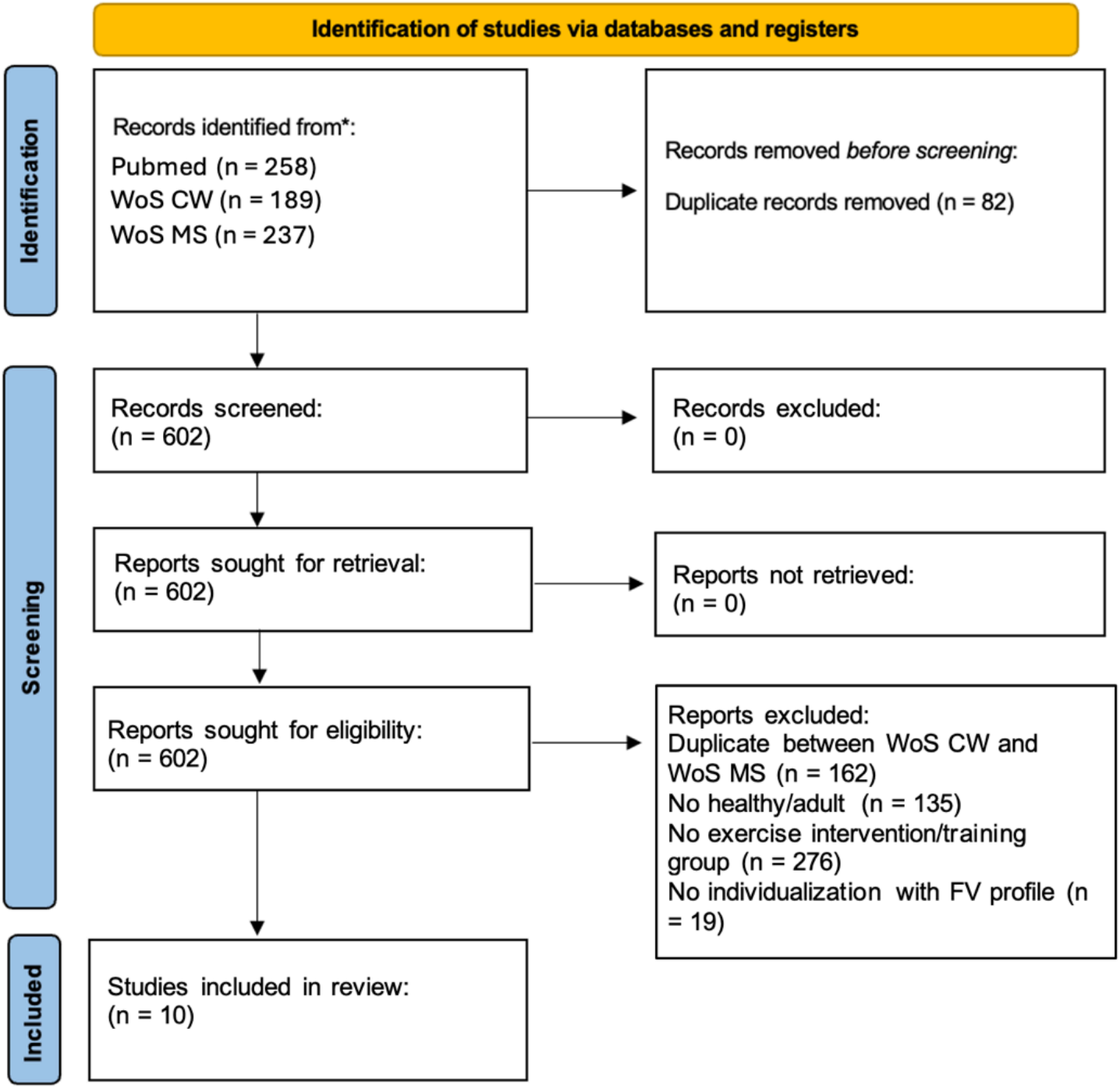


Figure 1

Flow of the literature search

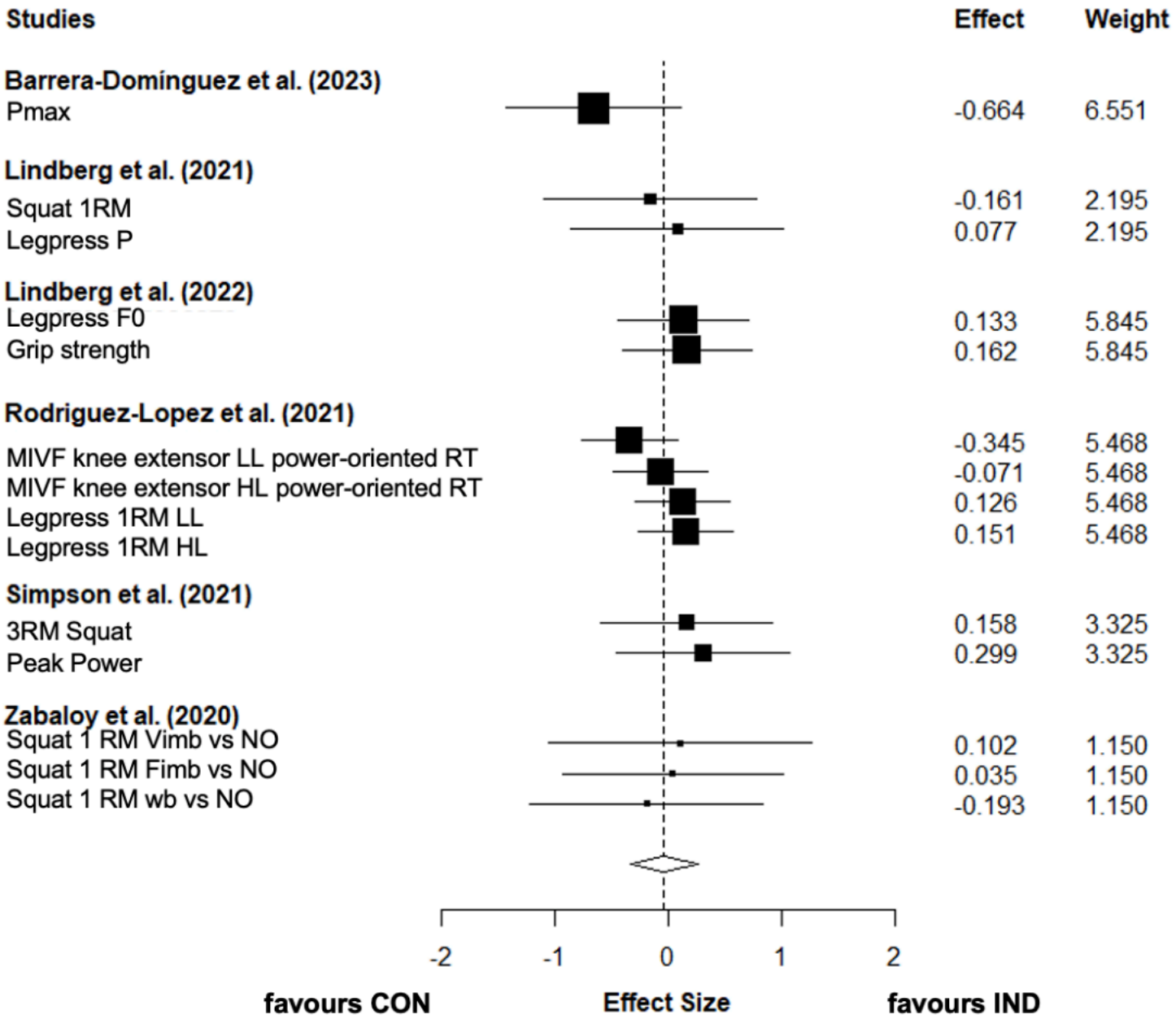


Figure 2

Effects of training individualization with the FV profile vs. non individualized group on strength

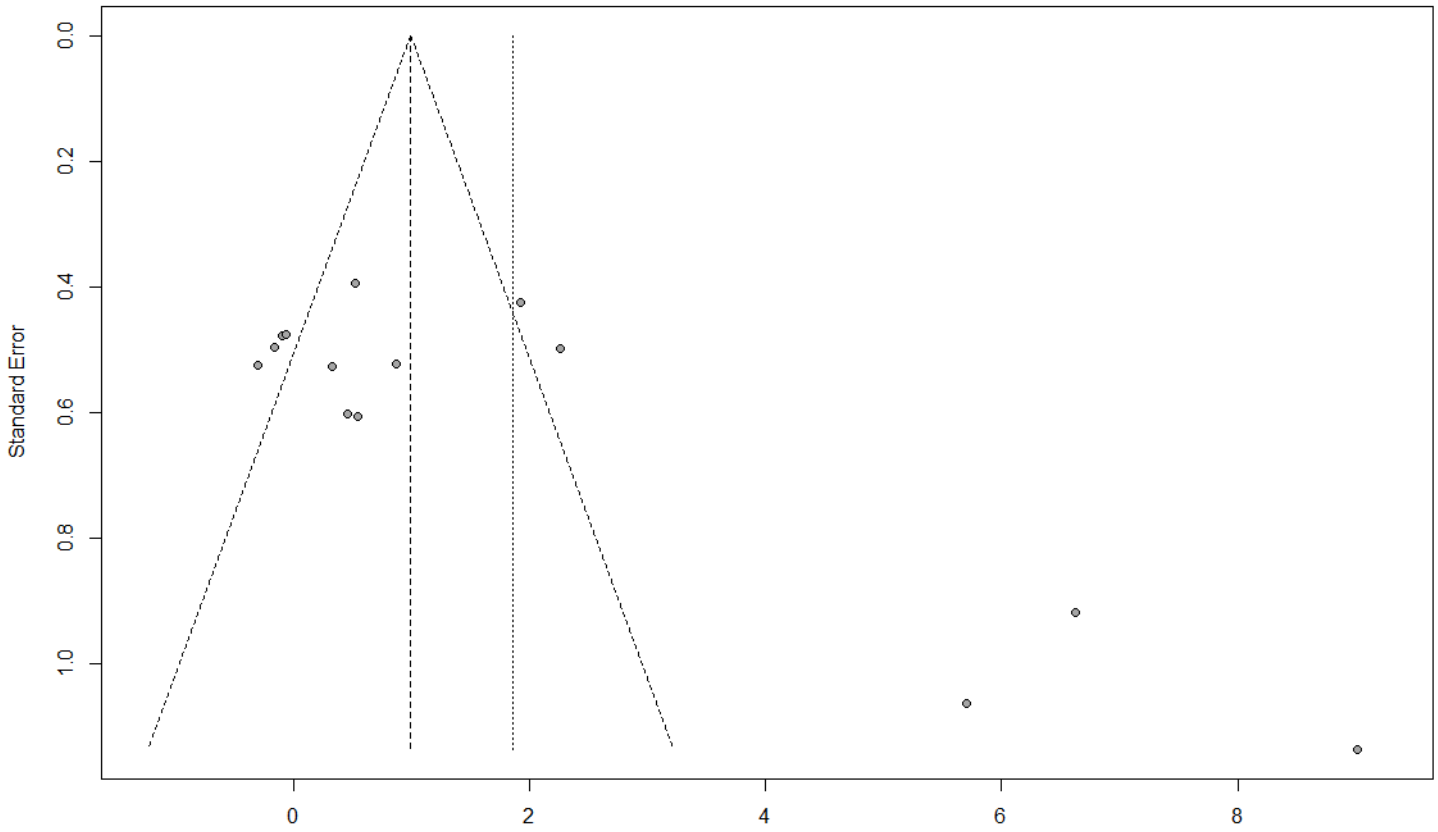


Figure 3

Funnel plot of the effects of individualization with the FV profile vs non individualized group on jump height

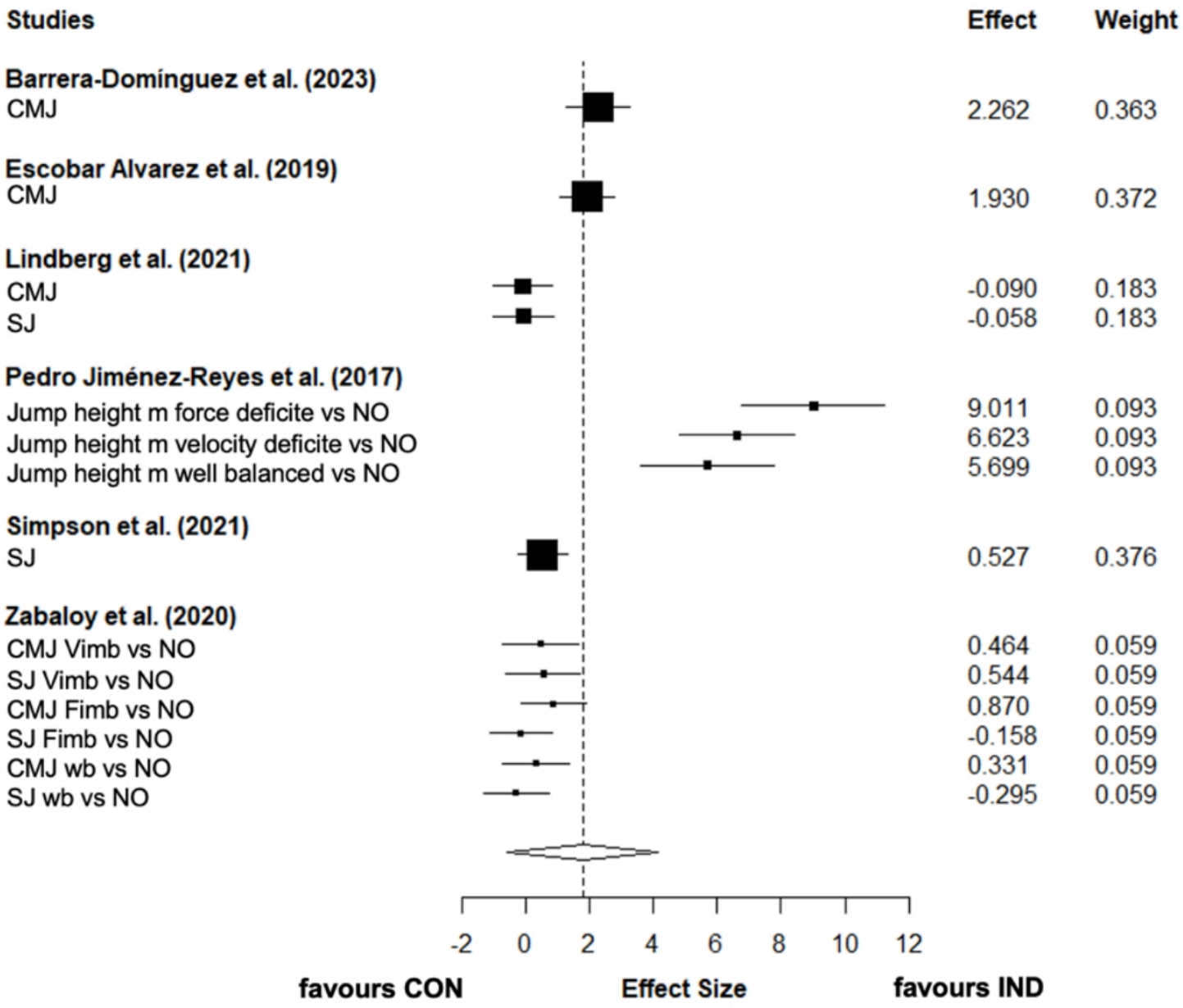


Figure 4

Effects of individualization with the FV profile vs. non individualized group on jump height.

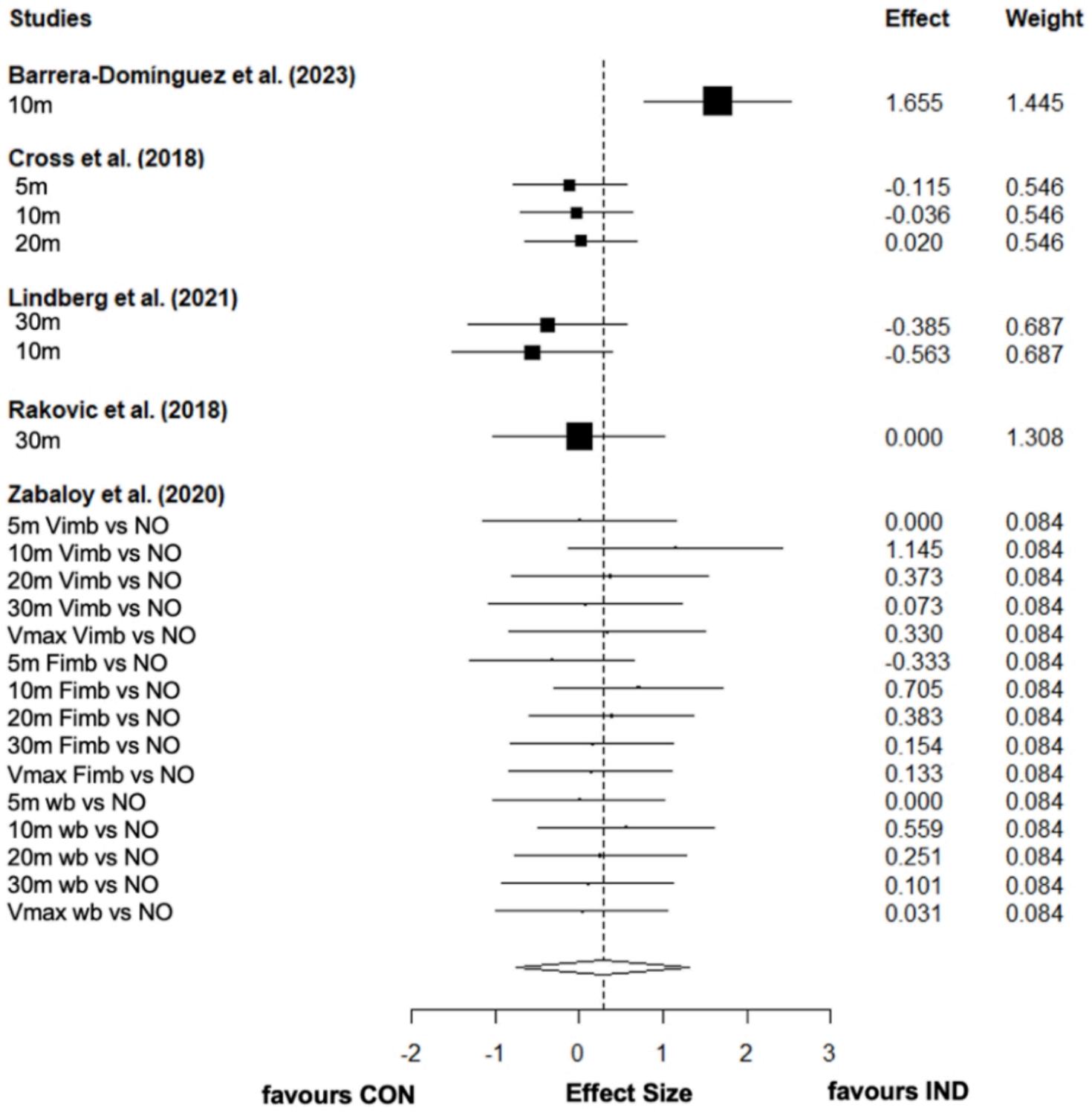


Figure 5

Effects of individualization with the FV profile vs. non individualized group on sprint time.

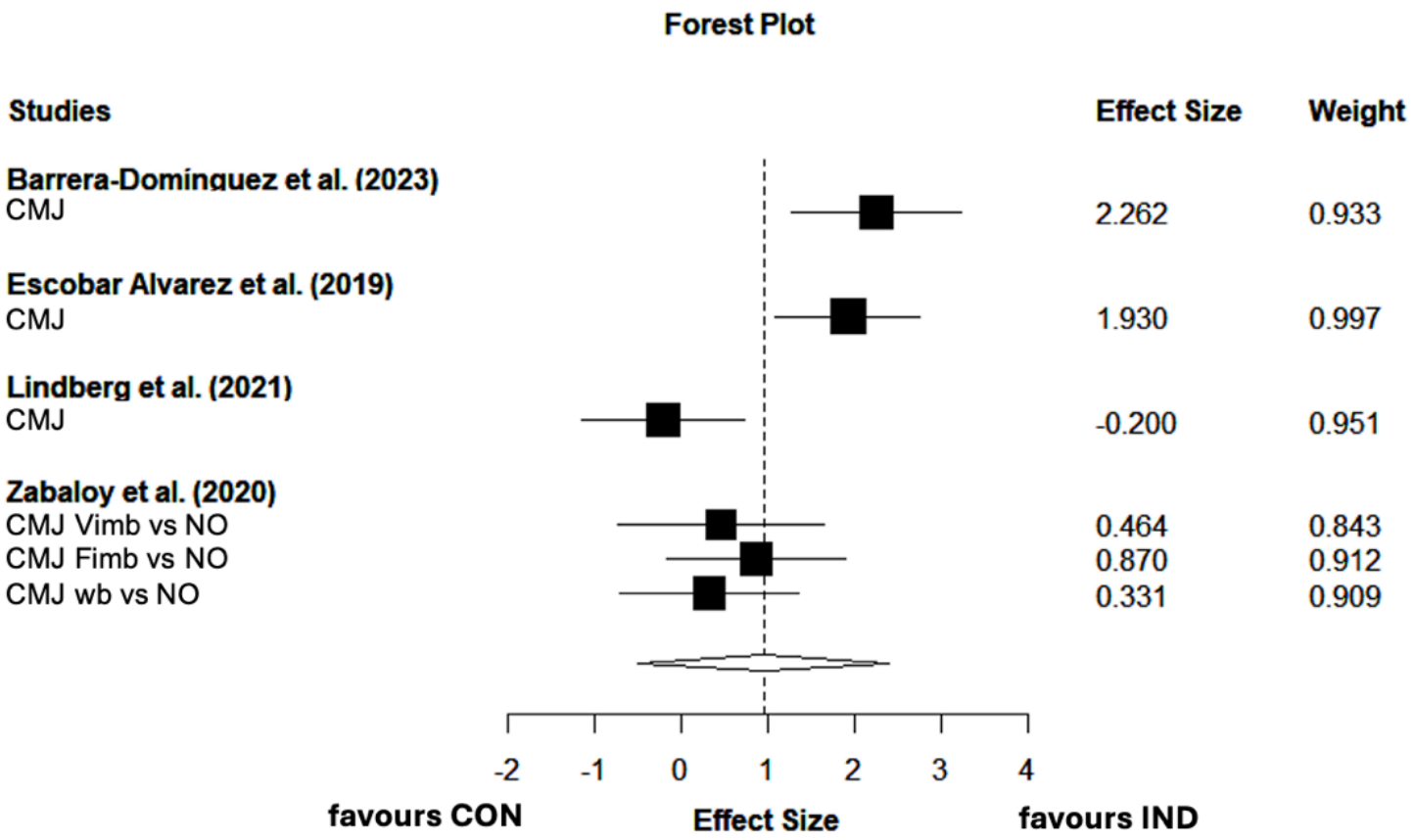


Figure 6

Effects of individualization with the FV profile vs. non individualized group on countermovement jump height.

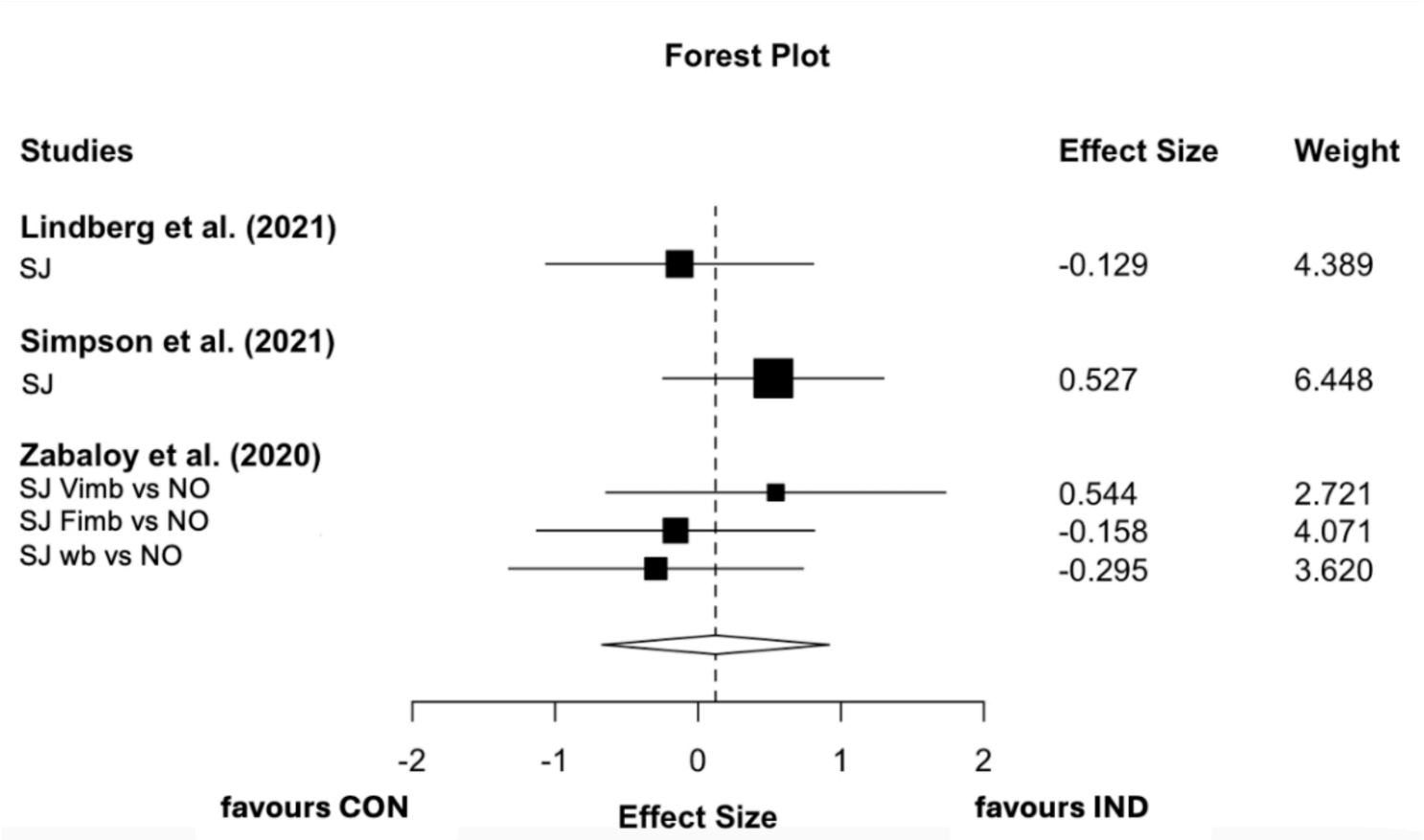


Figure 7

Effects of individualization with the FV profile vs. non individualized group on squat jump height.